

General questions / announcements.

□ Midterm?

□ Structure for the next weeks.

S M T W T F S

20 21 22 23 24 25

26 27 28 29 30 1 2

3 4 5 6 7 8

↑ Review

○ Lecture

○ Final

□ Change in Preclass → Postclass (8 onwards)
3pts → +3pts

MAYBE A FEW WORDS ON LINEAR ALGEBRA.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix};$$

$$Av = \lambda v, v \in \mathbb{R}^2.$$

• **Eigenvalues**: λ root of $\det(A - \lambda I) = 0$

• **Eigenvectors**: if $v \neq 0$, and $Av = \lambda v$

• **Eigenspace**: $\text{span}\{v_1, \dots, v_n\}$, v_i eigenvectors of λ .

• **Generalized eigenvector** $(A - \lambda I)^k v = 0$, $k > 1$.

• **determinant**: $\det A = ad - bc$

Example: Finding eigenvector:

$$\lambda \quad (A - \lambda I)v = 0 \quad \begin{bmatrix} a-\lambda & b & \vdots & 0 \\ c & d-\lambda & \vdots & 0 \end{bmatrix}, \quad v = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\det(A - \lambda I) = 0 \Rightarrow (a - \lambda)x + by = 0; \text{ if } b \neq 0$$

$$y = \frac{(\lambda - a)x}{b} \Rightarrow v = \begin{bmatrix} 1 \\ \frac{\lambda - a}{b} \end{bmatrix}$$

$$\det A = 0 \Leftrightarrow \begin{bmatrix} v_1 & v_2 \end{bmatrix}, \quad v_1 = \begin{bmatrix} a \\ c \end{bmatrix}, \quad v_2 = \begin{bmatrix} b \\ d \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad v_1 = \begin{bmatrix} a & b \end{bmatrix} \\ v_2 = \begin{bmatrix} c & d \end{bmatrix}$$

$\{v_1, v_2\}$ l.d.

$\{v_1, v_2, v_3\}$ in $\mathbb{R}^3$

$$v_1 + v_2 = v_3$$

$$\text{eig}(A) = 1, 1$$

$$\begin{array}{l} 0x + 0y = 0. \quad \dim 2 \\ \hline x + y = 0 \quad \dim 1. \end{array}$$

eigenspace(1) = dim 1 or 2.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1 \cdot \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{eigenspace} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Check: eigenvector has deg 1 of freedom

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \stackrel{\lambda=1}{(A-I)} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$y=0$, so x is free $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Lecture 07. 3.1 Linear Systems and Linearity Principle
3.2 Straight line solutions

Linear Systems

def. A system of differential equations

$$X'(t) = AX(t), \quad A \text{ is a } n \times n \text{ matrix.}$$

is called a linear system (with constant coefficients).

A is the coefficient matrix. $X' = A(t)X$

Goal: 2D systems.

$$\text{For } X(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \text{ and } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$X' = AX \quad \Leftrightarrow \quad \begin{cases} x' = ax + by \\ y' = cx + dy \end{cases}$$

Ex: Write as $X' = AX$.

$$x' = 42x + \pi y$$

$$y' = ex + 1y$$

$$X' = AX$$

$$A = \begin{bmatrix} 42 & \pi \\ e & 1 \end{bmatrix}$$

Natural questions:

• Equilibrium points?

$$X' = 0 \iff AX = 0$$

$$X = (0,0)? \checkmark$$

Any other? Maybe.

No: if $\det(A) \neq 0$, then exists A^{-1} $X = A^{-1}0 = 0$.

Yes if $\det(A) = 0$, then A is singular.

Examples:

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

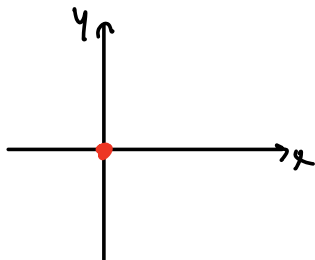
$$X' = AX$$

$$\det A = 12 - 9 = 3$$

$$x' = 2x + 3y$$

$$y' = 3x + 6y$$

so $(0,0)$ is the unique equilibrium.



$$B = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$$

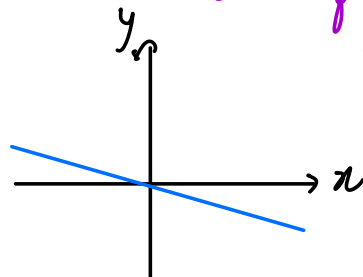
$$X' = BX$$

$$\det B = 12 - 12 = 0$$

$$x' = 2x + 4y$$

$$y' = 3x + 6y$$

so any point $x + 2y = 0$ is an equilibrium.



The Linearity Principle.

$$y' = py$$

Suppose $X' = AX$ is a linear system.

1. If $X(t)$ is a solution and k a constant, then

$kX(t)$ is also a solution, for any $k \in \mathbb{R}$.

2. If $X_1(t)$, $X_2(t)$ are two solutions, then

$X_1(t) + X_2(t)$ is also a solution.

Verify: 1. $(kX(t))' = kX'(t) = k \cdot AX(t) = A(kX(t))$. ✓

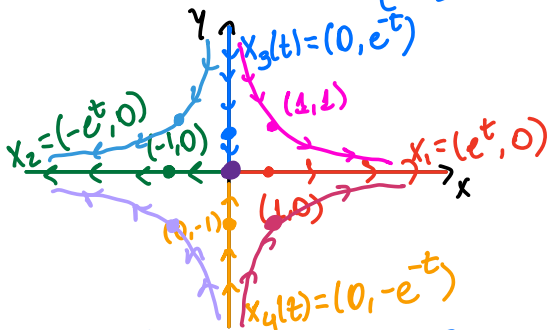
2. $(X_1 + X_2)' = X_1' + X_2' = AX_1 + AX_2 = A(X_1 + X_2)$ ✓

Ex: Sketch solutions of

$$\begin{cases} x' = x & \Rightarrow x(t) = c_1 \cdot e^t \\ y' = -y & \Rightarrow y(t) = c_2 \cdot e^{-t} \end{cases}$$

$$X_1(t) = \begin{bmatrix} c_1 \cdot e^t \\ 0 \end{bmatrix}, \quad X_2(t) = \begin{bmatrix} 0 \\ c_2 \cdot e^{-t} \end{bmatrix}$$

$$X(t) = c_1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + c_2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-t} \text{ is a solution.}$$



$$X_3(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \Rightarrow X_3(t) = \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$$

$$X_5 = X_1 + X_3 = \begin{bmatrix} e^t \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix} = \begin{bmatrix} e^t \\ e^{-t} \end{bmatrix}$$

$$X_5(0) = X_1(0) + X_3(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$X_6 = X_1 + X_4 = \begin{bmatrix} e^t \\ -e^{-t} \end{bmatrix}$$

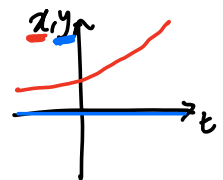
$$s = e^t.$$

$$e^{-t} = \frac{1}{s}. \quad \text{Curve } (s, \frac{1}{s})$$

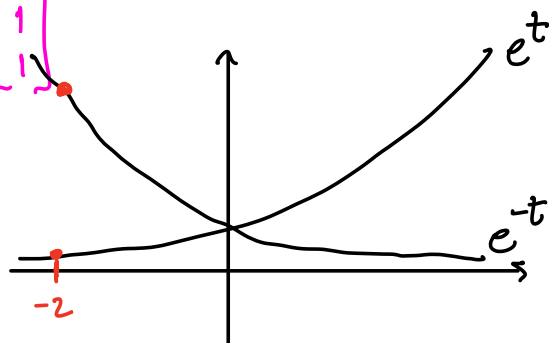
$$X_1(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^0 + c_2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-0}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \Rightarrow$$

$$X_1(t) = \begin{bmatrix} e^t \\ 0 \end{bmatrix} = \begin{bmatrix} x_1(t) \\ y_1(t) \end{bmatrix}$$



$$X_0 = X_1 + X_2 = \begin{bmatrix} e^t \\ 0 \end{bmatrix} + \begin{bmatrix} -e^t \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



THE GENERAL SOLUTION.

Suppose $X_1(t)$ and $X_2(t)$ are solutions for the initial value problem

$$X' = AX, \quad X(0) = X_0.$$

If $\{X_1(0), X_2(0)\}$ are linear independent, then
the general solution is given by $X(t) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-t}.$

$$X(t) = c_1 X_1(t) + c_2 X_2(t)$$

Ex: Solve the initial value problem

$$X' = AX \quad \text{with} \quad A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad X(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$X_1(t) = \begin{bmatrix} e^t \\ 0 \end{bmatrix}, \quad X_2(t) = \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$$

$$X(t) = c_1 X_1 + c_2 X_2$$

$$X(0) = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow X(t) = X_1 + X_2 = X_3$$

Ex: Undamped Harmonic Oscillator.

$$x'' = -x$$

a) Write as a system. by introducing $y = x'$.

$$\begin{aligned} x' &= y \\ y' &= -x \end{aligned}$$

b) Two solutions for ODE: $x'' + x = 0 \Rightarrow \lambda^2 + 1 = 0$
 $\lambda = \pm i.$

$$x_1(t) = \sin(t)$$

$$x_2(t) = \cos(t)$$

c) Two solutions for the system:

$$X_1(t) = \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} \quad X_2(t) = \begin{bmatrix} \cos(t) \\ -\sin(t) \end{bmatrix}$$

d) Verify linear independence:

$$X_1(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad X_2(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{linear independent}$$

Conclusion:

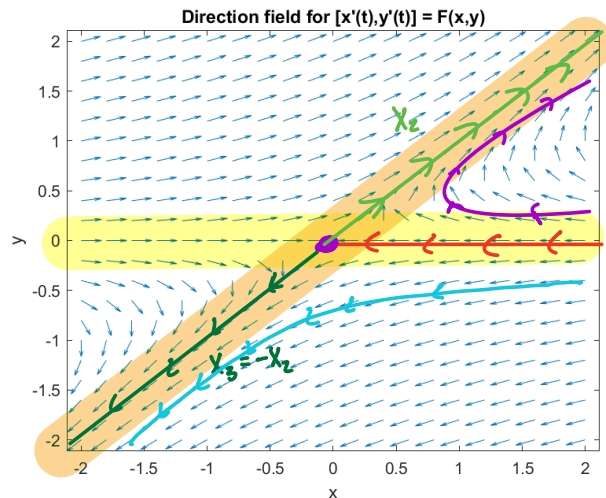
$$X(t) = c_1 \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} + c_2 \begin{bmatrix} \cos(t) \\ -\sin(t) \end{bmatrix} = \begin{bmatrix} c_1 \sin(t) + c_2 \cos(t) \\ c_1 \cos(t) - c_2 \sin(t) \end{bmatrix}$$

Straight-Line Solutions.

Ex.

$$\begin{cases} x' = 2y - x \\ y' = y \end{cases}$$

General solution:



$$y(t) = d e^t, \quad x' = -x + 2y = -x + 2d e^t$$

$$x_h = c \cdot e^{-t} \quad x_p = A e^t; \quad x_p' = A e^t.$$

$$A e^t = -A e^t + 2d e^t \Rightarrow 2A = 2 \Rightarrow A = d.$$

$$x(t) = c_1 e^{-t} + e^t, \quad c_1 = 1.$$

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} e^{-t} + e^t \\ e^t \end{bmatrix} \quad \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} c e^{-t} + d e^t \\ d e^t \end{bmatrix}$$

$$= c \cdot e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

What happens when $d=0$?

$$x_1 = \begin{bmatrix} e^{-t} \\ 0 \end{bmatrix}$$

What happens when $c=0, d=1$?

$$x_2 = \begin{bmatrix} e^t \\ e^t \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

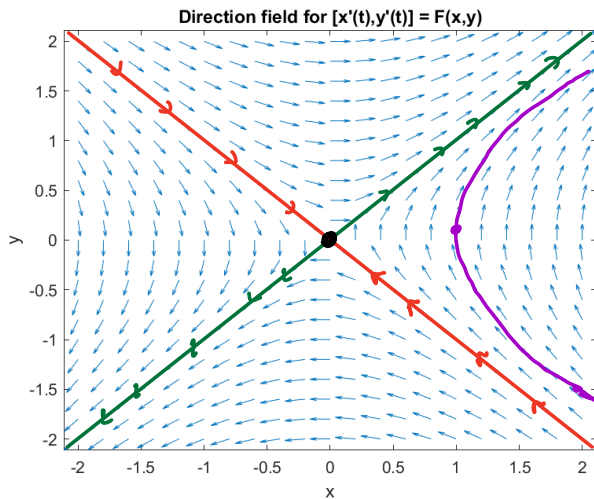
Ex

$$\begin{cases} x' = y \\ y' = x \end{cases}$$

General solution: (2nd order)

$$x_2(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x_1(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$x' = y \Rightarrow x'' = y' = x \Rightarrow x'' - x = 0 \Rightarrow \lambda^2 - 1 = 0 \\ \lambda = \pm 1.$$

$$x_1(t) = c_1 e^t, \quad x_2(t) = c_2 e^{-t}$$

$$y_1(t) = c_1 e^t, \quad y_2(t) = -c_2 e^{-t}$$

$$X_1(t) = c_1 \begin{bmatrix} e^t \\ e^t \end{bmatrix}, \quad X_2 = c_2 \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

$$= c_1 e^t \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{v_1}$$

$$X_2 = c_2 e^{-t} \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{v_2}$$

Finding straight-line solutions:

$$Av = \lambda v$$

Suppose λ is an eigenvalue, v associated eigenvector.

if $x(t) = v \cdot e^{\lambda t}$,

$$x'(t) = v \cdot \lambda e^{\lambda t} = (\lambda v) \cdot e^{\lambda t} = Av \cdot e^{\lambda t} = A(v e^{\lambda t}) = A x(t)$$

so $x_1(t), x_2(t)$ can be written in terms of the eigenvalues and eigenvectors of A .

Back to the example:

$$\begin{cases} x' = y \\ y' = x \end{cases}$$

$$x' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} x$$

$$x(t) = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \text{ eigenvalues: } \lambda = \pm 1. \quad x(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}.$$

$$\underline{\lambda = 1}: \begin{bmatrix} -1 & 1 & : & 0 \\ 1 & -1 & : & 0 \end{bmatrix} \rightarrow v = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } -x + y = 0 \Rightarrow y = x, v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\underline{\lambda = -1}: \begin{bmatrix} 1 & 1 & : & 0 \\ 1 & 1 & : & 0 \end{bmatrix} \rightarrow x + y = 0 \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Ex: Which one is straight line solutions of

$$\begin{cases} x' = 2x + 3y \\ y' = -4y \end{cases} \quad A = \begin{bmatrix} 2 & 3 \\ 0 & -4 \end{bmatrix}$$

a) $X(t) = \begin{bmatrix} -e^{-4t} \\ 2e^{-4t} \end{bmatrix} = e^{-4t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \checkmark$

c) $X(t) = \begin{bmatrix} 0 \\ e^{2t} \end{bmatrix} = e^{2t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

b) $X(t) = \begin{bmatrix} -e^{-4t} \\ -2e^{-4t} \end{bmatrix} = e^{-4t} \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

d) $X(t) = \begin{bmatrix} 1 \\ e^{2t} \end{bmatrix} \times$

$\lambda = -4$

$(A - \lambda I)x = 0$

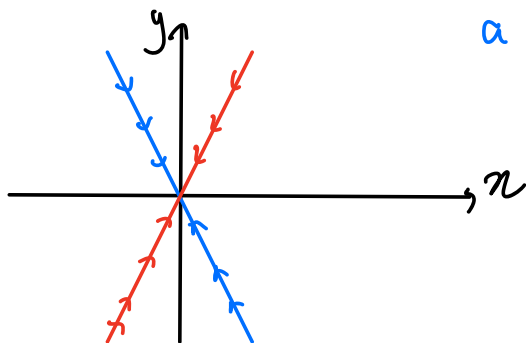
$$\begin{bmatrix} 2 - (-4) & 3 & : & 0 \\ 0 & -4 - (-4) & : & 0 \end{bmatrix} = \begin{bmatrix} 6/3 & 3/3 & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$$

$2x + y = 0$

a) $2 \cdot (-1) + 2 = -2 + 2 = 0 \checkmark$

b) $2 \cdot (-1) + (-2) = -4 \neq 0 \times$

a) $\begin{bmatrix} 2 & 3 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2+6 \\ -8 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix} = -4 \begin{bmatrix} -1 \\ 2 \end{bmatrix} \checkmark$, i.e., $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ is eigenvector for $\lambda = -4$.



a b

$$e^{-4t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Motivation: where is this coming from?

$$\begin{array}{ll} x' = ax & \text{vs} \quad X' = AX \\ x(t) = e^{at} & X(t) = e^{At} \end{array}$$

Exponential matrices.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Relation with Euler's Formula

$$e^{it} = \cos(t) + i\sin(t)$$

$$\begin{aligned}
 e^{it} &= 1 + (it) + \frac{(it)^2}{2} + \frac{(it)^3}{3!} + \frac{(it)^4}{4!} + \dots \\
 \cos(t) &= 1 - \frac{t^2}{2} + \frac{t^4}{4!} - \dots \\
 i\sin(t) &= (it) - \frac{i(t)^3}{3!} + \frac{i(t)^5}{5!} - \dots
 \end{aligned}$$

$i^2 \cdot t^2 = -t^2$
 $i^4 \cdot t^4 = t^4$
 $-i = i^3$
 $i = i^5$

$$e^{it} = \cos(t) + i\sin(t)$$

What about matrices?

$$e^{At} = \mathbb{I} + (At) + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \frac{(At)^4}{4!} + \dots + \frac{(At)^n}{n!} + \dots$$

Next: Finish matrix exponential + Real & Complex case.

the end.